

MULTIMODAL BEHAVIORAL USER AUTHENTICATION BASED ON KEYSTROKES AND MOUSE MOVEMENTS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Behavioral biometrics are novel methods for authentication, leveraging unique characteristics such as non-repudiation and unforgeability. Behavioral biometrics encompass various information derived from various human actions such as walking patterns (gait), voice, handwriting, and user activity logged on peripheral devices like keyboards, mice, smartphone sensors, webcams, and others. This study aims to design and implement a multi-model authentication framework that integrates keystroke dynamics and mouse movements for robust user authentication in practical applications. A customized software-based keystroke and mouse coordinated logging tool is developed to collect the user's keystrokes and mouse movements across different steps of page navigation. This custom tool encompasses a login page, which serves as the primary focus for our authentication system implementation. Temporal and statistical features are derived from the collected data and are fed into a novel user registration architecture based on Gated Recurrent Unit (GRU) for analysing keystrokes and Support Vector Machine (SVM) for combining the output probabilities of GRU with analysed mouse movements. Upon training the system on the University of Buffalo shared keystroke dataset comprising 148 users, the model demonstrated strong performance across selected evaluation metrics. Additionally, our system can be scaled up to a high usability real-time authentication system which has practical significance especially in corporate environments.

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CHAPTER 1

INTRODUCTION

In computer security, ensuring robust user authentication mechanisms is essential to safeguarding sensitive data and mitigating potential security threats. Traditional authentication methods, such as passwords and tokens, have demonstrated vulnerabilities by being exposed to various forms of cyber attacks, underscoring the need for more resilient and unforgeable alternatives. In response to these challenges, behavioral biometrics has emerged as a promising approach to authentication, leveraging unique behavioral patterns inherent in users' interactions with computing devices, such as their intrinsic attributes like voice or typing patterns.

1.1 MOTIVATION

Computer Security is an ever-expanding area of research due to the ongoing competition between security systems and evolution of potential adversaries. Since thousands of transactions are processed everyday and require to be completed without much exposure, highly secure protocols to be developed to encrypt and protect the underlying systems [16]. While exploring more advanced methods of authentication, it is discovered that behavioral biometric attributes such as touch dynamics, keystroke dynamics and gait recognition could be effectively and efficiently used to authenticate individuals. These solutions are promising due to being able to verify the authenticity of individuals in an implicit

manner [2].

Many solutions employ static authentication, that is, they authenticate users at only the first step. Therefore, such solutions are inefficient [1] due to the problem of session hijacking when the user fails to log out of the system before leaving the premises. Furthermore, utilization of just a single modality reduces the accuracy of the system by a large margin because of the low amount of samples needed to crack the users' identity and in turn limited data to securely tag a person [22]. By combining different behavioral modalities in a systematic manner, a more nuanced understanding of the user profile is obtained, paving the way for enhanced accuracy and reliability.

1.2 BACKGROUND

1.2.1 History of Authentication

Authentication is the process of verifying the digital identity of users [12]. It is the process of verifying the identity of a user, system, or device, often as a prerequisite to allowing access to resources in an information system. The phenomenon of authentication as demonstrated in Figure 1.1¹, has a rich history. ranging from a simple card or password-based authentication to more complex protocols involving location, biometrics, and behavioral dynamics as we know today. The evolution of authentication has been marked by significant advancements and changes in

¹Evolution of Authentication: <https://d3aafpijpsak2t.cloudfront.net/images/Blog-Images/Authentication-blog-imgv2.png>

response to evolving technology and emerging threats. Starting from the mid-1960s, authentication methods have continually evolved to address new challenges and improve security.

EVOLUTION OF AUTHENTICATION

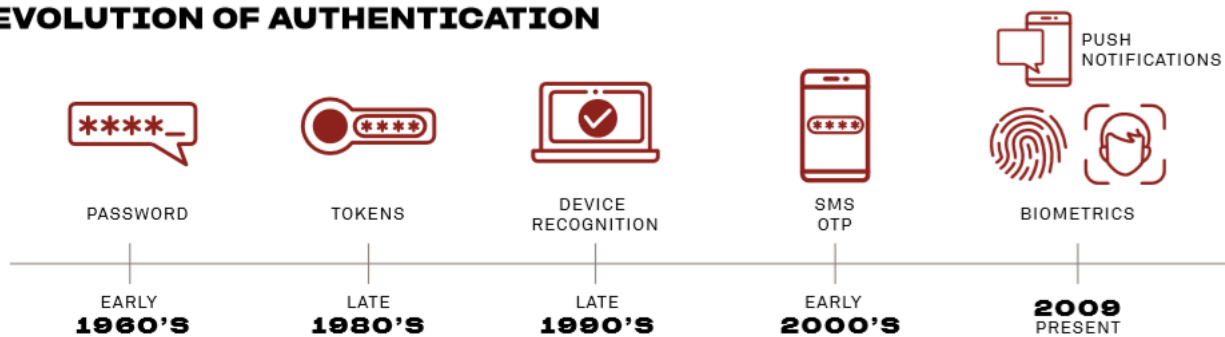


FIGURE 1.1: Evolution of Authentication

In the mid-1960s, passwords and hash techniques emerged as the primary method of authentication. Passwords, initially simple character strings, were hashed using cryptographic algorithms such as MD5 or SHA-1 for storage and comparison. However, these methods were susceptible to brute-force attacks and dictionary attacks due to weak password policies and inadequate hashing techniques.

Token-based and device-based authentication gained popularity in the 1960s and 70s, offering an additional layer of security beyond passwords. Tokens, such as smart cards or physical tokens, generated one-time passwords (OTPs) or cryptographic keys for user authentication. These methods enhanced security by requiring possession of a physical device in addition to knowledge of a password. However, they were still vulnerable to theft, loss, or duplication of tokens, leading to potential security breaches.

During the 1980s and mid-1990s, OTP-based and CAPTCHA-based authentication mechanisms emerged to combat increasingly sophisticated attacks.

OTPs provided a temporary code that users entered alongside their passwords for authentication, adding an extra layer of security. CAPTCHAs, on the other hand, were introduced to distinguish between humans and automated bots, preventing automated attacks such as brute-force or spamming.

The early 2000s witnessed the emergence of two-factor authentication (2FA) and subsequent multi-factor authentication (MFA) systems, addressing the limitations of single-factor authentication methods. 2FA combined something the user knows (password) with something they have (token or device), significantly enhancing security by requiring multiple forms of verification. MFA expanded on this concept by incorporating additional factors such as biometric data (something they are) or location-based authentication, further strengthening security measures.

In the 2010s to the present, the emergence of biometrics revolutionized authentication methods by leveraging unique physical or behavioral traits for user verification. Biometric authentication methods, including fingerprint recognition, facial recognition, iris scanning, and voice recognition, offer a high level of security and user convenience. These methods utilize encryption techniques to protect biometric data stored in databases, along with sophisticated algorithms for pattern recognition and authentication. Biometric authentication by itself has its own history, dating back to the simple techniques used by the Babylonian empire to the more advanced methods used in modern biometrics.

1.2.2 History of Biometric Authentication

The history of biometrics, as demonstrated in Figure 1.2², traces back to ancient civilizations, with the Babylonian empire among the first to employ basic biometric techniques. However, the formal development of biometric identification systems began in the 1800s, marked by Alphonse Bertillon's method of classification which based on measurements taken on the body. Bertillonage, as it was known, laid the groundwork for using physical characteristics for authentication, despite its eventual shortcomings due to human variability.

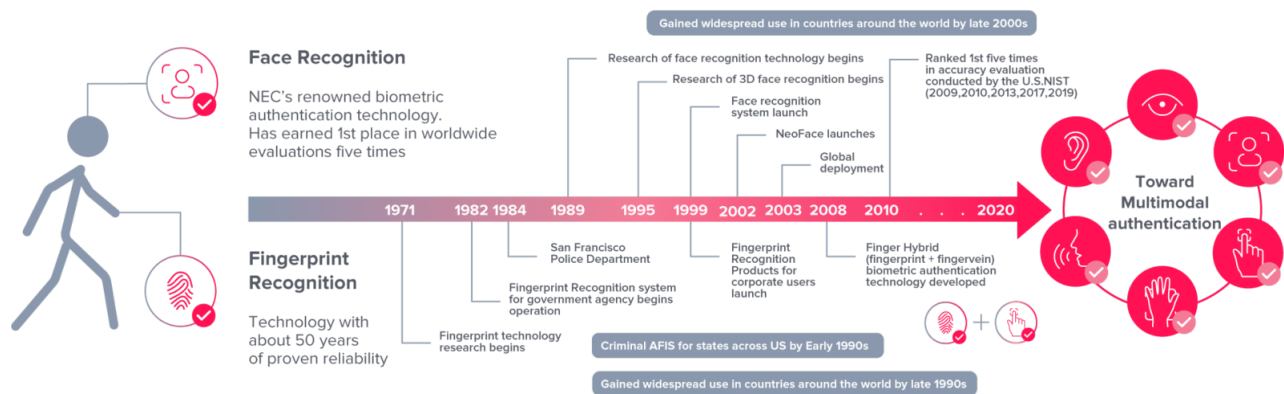


FIGURE 1.2: Evolution of Biometric Authentication

In the late 1800s, the focus shifted to fingerprints, with Sir Francis Galton proposing a classification system based on fingerprint minutiae. This led to the collaboration between Galton and Sir Edward Henry, resulting in the Henry Classification System, which became the standard for fingerprint identification. By the early 1900s, fingerprint databases were established, laying the foundation for modern biometric authentication.

²Evolution of biometric authentication: <https://recfaces.com/wp-content/uploads/2020/12/history-of-biometrics.png>

The 20th century witnessed significant advancements in biometric technology, including the exploration of iris patterns by ophthalmologist Frank Burch and the automation of fingerprint identification by the FBI in the late 1960s. The 1970s saw the emergence of facial recognition research, although initial methods relied on manual computations of facial markers.

The 1980s and 1990s saw further innovations, with the development of hand geometry and vascular pattern recognition technologies. Additionally, research efforts expanded to include speaker recognition, leading to the formation of the NIST Speech Group and advancements in acoustic speech analysis.

The turn of the century marked a new era for biometrics, with the launch of large-scale tests such as the Face Recognition Vendor Test (FRVT) and the establishment of academic programs focused on biometric systems. The 2000s also saw widespread adoption of biometric solutions in various sectors, from government initiatives like US-VISIT to commercial applications like MasterCard's Biometric Card.

In recent years, advancements in biometric technology have accelerated, with the introduction of features like Touch ID and voice assistants. The integration of biometrics into everyday devices and applications has paved the way for a passwordless society, with experts predicting further innovations such as heart-rate and gait detection in the near future.

As we enter the 2020s, the landscape of biometrics continues to evolve rapidly, with the emergence of various behavioral biometrics. Newer modalities like keystroke dynamics and mouse movements have gained prominence. Furthermore, multi-factor authentication has evolved into multi-modality

authentication, combining different biometric modalities for enhanced security and user experience.

1.3 PROBLEM DEFINITION

1.3.1 Problem Statement

The identified problem is to develop a multi-modal authentication system which can detect and identify users by using a machine learning model which is trained based on behavioral data pertaining to their unique keystrokes and mouse movements. The system allows for registration of unique users by capturing and analyzing their keystroke and mouse movement patterns, establishing individual profiles similar to unique digital fingerprints. Using these registered profiles, the authentication process is to detect the presence of an ‘imposter’, i.e., one whose keystroke and mouse movement patterns deviate from those expected of the claimed user. The information of a potential imposter detection would then be then used for further actions such as restricting access or issuing alerts to the original user through a more well-established secure channel.

1.3.2 Justification

With the increase in activities like password-hacking, session-hijacking and potential vulnerabilities inherent in other secondary authentication methods present, there is an increased motivation to engineer an authentication system which is tailored to individual traits, while simultaneously requiring little to no

infrastructure to develop the same. This is where the benefit of using keystrokes and mouse movement data for user profiling and authentication is brought into the picture. With just a keyboard and a mouse, the user would then be able to setup a unique pseudo-fingerprint based on his own typing and mouse usage patterns, avoiding the need for additional hardware to employ finger or face recognition systems.

1.3.3 Scope

The scope of the project is to develop a multi-modal keystrokes and mouse-movements authentication system tailored to corporate environments where session hijacking and spoofing are more common. The main output of the project is a probabilistic report that assesses the likelihood of an individual being an imposter. The probabilistic nature stems from the inherent slight variations in behavioral biometrics with each instance, unlike passwords which possess an inherent deterministic nature. To complement this ability, the system incorporates certain probabilistic threshold values which are compared against the calculated probability values of the individual being an imposter. In contrast to exact match for passwords, these thresholds are used to compare and classify the likelihood of an individual being an imposter.

1.4 ORGANIZATION OF THE REPORT

The report is organized into six main sections, each contributing to a comprehensive understanding of the research conducted. Chapter 2, the Literature

survey, reviews relevant papers pertaining to the identified problem, highlights existing gaps in the literature, and delineates the research objectives. In Chapter 3, the Proposed Methodology, the methodology employed in the project work is detailed, with sub-sections tailored to the specific aspects of the research. Chapter 4, Experimental results and performance analysis, provides a thorough description of the dataset used, outlines the hardware and software specifications of the experimental ecosystem, presents the experiments conducted, and offers a comprehensive analysis of the performance results obtained. Chapter 5, Social Impact & Sustainability, delves into the broader societal implications and sustainability considerations of the proposed multi-modal authentication system. Finally, Chapter 6, Conclusions and future work, synthesizes the findings of the study, outlines conclusions drawn from the research outcomes, and suggests avenues for future research and development in the field.

CHAPTER 2

LITERATURE SURVEY

The section highlights the existing work in the domain of behavioral biometrics to provide security for the existing system [24], with a main focus on keystrokes, mouse dynamics, and other sensory inputs like touch and gyroscopic sensors used in smartphones. Through these biometrics, the systems attempted to distinguish between normal inputs and imposter inputs through the creation of a *virtual fingerprint* pertaining to their specific inputs [15]. In addition, the absence of a need for any other sensors (especially in the case if keystroke and mouse movement dynamics where you need nothing more than a keyboard and a mouse) has diminished the complexity of the underlying hardware architecture, and subsequently its cost factor [7].

2.1 EXISTING LITERATURE

This Literature Survey is divided based on the behavioral data they concentrate on. This includes:-

- Keystroke Dynamics
- Mouse Dynamics
- Related work on other behavioral authentication systems (E.g.: Touch Dynamics on Smartphones)

2.1.1 Keyboard dynamics

The realm of keystroke dynamics delves into the intricate nuances of users' typing habits, encompassing both temporal and spatial aspects. Extensive research indicates that seasoned typists exhibit consistent keying patterns [25], which remain relatively unchanged even under intentional external influences. Consequently, keystroke dynamics emerges as a promising avenue for biometric identification. Numerous thorough reviews, such as those conducted by Ali et al. [3], provide comprehensive insights into this field. While the keystroke process primarily revolves around the timing of key presses and releases, variations may arise based on factors like keyboard layout and user behavior. Consequently, many distinctive typing behaviors extend across different devices, akin to a pianist's unique touch on a familiar instrument.

- The authors of [10], proposed a simple and secure authentication scheme which aims to authenticate users by combining their typing rhythm and text password, employing a statistical approach in a two-phase policy encompassing registration (involving data capture, feature extraction, and learning) and log-in (utilizing keystroke dynamics features such as press time, inter-key time, and total time). The authors evaluated their method on a dataset of 100 users.
- One of the pioneering endeavors in the realm of free script analysis was pioneered by the authors of [9]. Their approach involved analyzing free-form text for authentication purposes through a tailored interface, prompting users to type a passage of approximately 700–900 words. The analysis revolves around studying the typing rhythm, leveraging timing

data from individual keystrokes, and comparing n-graphs of consecutive text entries. The setup entails capturing keystroke data using a number pad to train the system. Following the training phase, the system undergoes testing, where it captures the user's keystroke dynamics once again and compares it with the user keystroke profile established during training. Upon successful authentication, the user is prompted to input a password.

- Another statistical method, relying on mutual feature vectors, is employed for authentication, as proposed by the authors of [26]. In the training phase, keystroke features of the genuine user are regularly generated and recorded in the database. Then, during authentication of current users, previously stored feature vectors and distortions are computed. The experiments conducted in this study involved multiple passwords typed by the user. However, it's important to note that this approach focuses on short fixed texts, without accounting for potential changes in user behavior over time.
- The model proposed by the authors of [27] suggests that the intervals between typing different characters on the keyboard may be attributed to factors such as unfamiliar words or spelling, leading to pauses. Additionally, they argue that cognitive factors unique to each individual contribute to these patterns. Consequently, various machine learning techniques can be employed in authentication systems to verify users. However, it's worth noting that the experiments conducted were limited to simulated environments, raising questions about its applicability to real-time authentication. Furthermore, the training time required for the system is not addressed. In addressing the issue of biometric permanence in biometric systems, the authors of [17] propose a timing-based

authentication system using keystroke dynamics. Their work focuses on studying how biometric signatures change over time.

- The authors of [11] highlighted an important observation regarding keystroke information, suggesting that existing methods need customization to accommodate keyboards other than the traditional PC keyboard. This underscores the need for adaptable approaches to effectively capture keystroke dynamics across diverse input devices.

2.1.2 Mouse Dynamics

Mouse dynamics are similar to keystroke dynamics, the difference is that here the mouse movements and actions are used to regulate whether a user is reliable or not. Typically, mouse movements, gestures, clicks, and their timing information are collected in authentication systems based on mouse dynamics [4]. The method has been applied to both static authentication and active authentication, yielding diverse outcomes. In static authentication, mouse movements are collected for a predefined task during the training phase, forming a profile for the legitimate user. During testing, the user repeats the same task, and the authentication system verifies their authenticity. Conversely, in active user authentication [30], the task is not predetermined, allowing the scheme to authenticate the user across various tasks and activities.

- In their work [18], the authors introduced a graphical authentication system named Mouse Lock, where users select a set of images and move them in a specific direction and sequence to form a password. The timing data of

consecutive image movements is utilized to determine the authenticity of the user. However, increasing the number of movements excessively to enhance security could make the login process cumbersome. Additionally, this method's intrusive nature is often viewed as a drawback in behavioral authentication.

- Utilizing mouse gesture analysis for static authentication during login presents a promising avenue in biometric systems. Building on this, [19] conducted a study involving 39 users to refine the mouse gesture method. In this scheme, users are required to draw gestures eight times for analysis. The system is trained using 14 data points samples, and during testing, deviations from these data points are measured to authenticate the user. However, to adapt this method for non-intrusive continuous authentication, modifications are necessary. One notable limitation is the questionable security aspects of this approach.
- Proposing a novel approach, [13] suggests that combining mouse dynamics with non-behavioral modalities such as eye movement could lead to improved performance. In their study, eye statics are recorded at a lower frequency. However, despite this limitation, the devised scheme demonstrated a high flaw measure.

2.1.3 Other related work in Behavioral Dynamics

- The authors of [6], proposed a continuous authentication system for mobile devices that uses software keyboard interactions as a biometric. The system

builds a biometric profile for each user based on their touch pressure, location, and timing.

- The authors of [21], proposes a multi-modality based framework for continuous user authentication on smartphones, called MMAAuth. MMAAuth integrates heterogeneous information of user identity from multiple modalities (e.g., motion movement pattern, touch dynamics, usage context) to perform more accurate authentication.

Table 2.1 summarizes the important work on behavioral dynamics derived from the above literature survey.

TABLE 2.1: Existing Literature on Behavioral Dynamics

S.No.	Author (Year)	Features	Dataset	Limitation or Remarks
1	Haque et al. (2015)	Typing rhythm, text password	Dataset of 100 users	Requires further evaluation in real-time scenarios
2	Gunetti et al. (2005)	Free text analysis, timing information	Custom dataset	Intrusive method, potential usability issues
3	Wang et al. (2012)	Mutual feature vectors, distortion calculation	User-generated dataset	Limited consideration for user behavior changes over time
4	Chang et al. (2016)	Pause time intervals, cognitive factor	Simulated environments	Lack of clarity on real-time translation, training time not discussed
5	Pisani et al. (2015)	Timing phase system, biometric permanence	Dataset not specified	Limited discussion on system scalability
6	Kang et al. (2015)	Keystroke adaptation for non-traditional keyboards	Not specified	Potential need for customization for different keyboard types
7	Revett et al. (2008)	Mouse lock graphical authentication	Image movement timing	Intrusive method, potential usability issues
8	Sayed et al. (2013)	Mouse gesture analysis	Data from 39 users	Security aspects need improvement, non-intrusive modification required
9	Kasprowski et al. (2018)	Mouse dynamics combined with eye movement	Eye statics recorded in low frequency	Potential improvement in performance with eye movement integration
10	Dee et al. (2019)	Software keyboard interactions as biometric	Mobile device interactions	Continuous authentication system, potential scalability concerns
11	Shen et al. (2022)	Multi-modality based authentication framework	Smartphone usage context	Integration of heterogeneous information, accuracy improvements

2.2 RESEARCH GAPS IDENTIFIED

Combining Keystroke Dynamics (KD) and Mouse Dynamics (MD): While both KD and MD have been studied extensively as standalone methods, there is little research focusing on their combined use. This gap suggests an opportunity for future studies to investigate the benefits, challenges, and performance implications of integrating KD and MD into a single, cohesive authentication

framework.

User-Specific Models: Using one model for each user in a personalized approach to keystroke dynamics and mouse movements authentication offers several advantages. Firstly it is much simpler in the sense of Machine Learning to train models which perform the user/imposter identification compared to a single model approach which leads to the much harder user/user classification. Additionally, a single model for a user is a direct answer to the scalability problem, preventing corruption of existing user signatures upon the creation of a newer model.

Integration with Existing Systems: Most of the works on the literature focus on a theoretical understanding of the discriminability and the performance of keystroke and mouse-movement dynamics and lacks comprehensive studies on the integrability of behavioral biometric methods into practical, usable security solutions. Setting the right authentication algorithms, ensuring consistency in the data-collection in real-time environments, and ensuring scalability when integrating user profiles can be some of the crucial challenges in practical integration.

2.3 RESEARCH OBJECTIVES

The main objective of the project is to develop a system that creates a user profile, or *virtual fingerprint* based on the user data pertaining to both keystrokes and mouse movements. Previous studies have either focused on keystroke and

mouse-movement dynamics separately, or they have combined keystrokes or mouse movements with other physiological biometrics (Eg: Mouse + Eye movement explored in the above related works). Additionally, the other studies have not explored the practical integration of these combination to multiple modalities into a scalable system. Our main contributions to this domain are

- Explore the practical integration of keystroke and mouse-movement dynamics into a functional application, which stores the user keystrokes and mouse movements into logically separate user profiles/signatures.
- Investigate into a novel approach which combines a user's keystrokes and mouse movements to create a personalized *virtual fingerprint* which could be later used for user verification and authentication.

CHAPTER 3

PROPOSED METHODOLOGY

The Proposed Design, as indicated in Figure 3.1 and 3.2, combines keystroke dynamics (KD) and mouse dynamics (MD) into a single unified authentication framework. By integrating both modalities, the system aims to leverage the strengths of each method while compensating for their individual limitations. Additionally, the implementation adopts a user-specific modeling approach, where personalized models are created for each user based on their unique behavioral patterns. This approach enhances reliability and accuracy by tailoring the authentication model to the specific characteristics of each user. Furthermore, using individual models for each user helps to mitigate the risk of data contamination and ensures robustness in the overall authentication system.

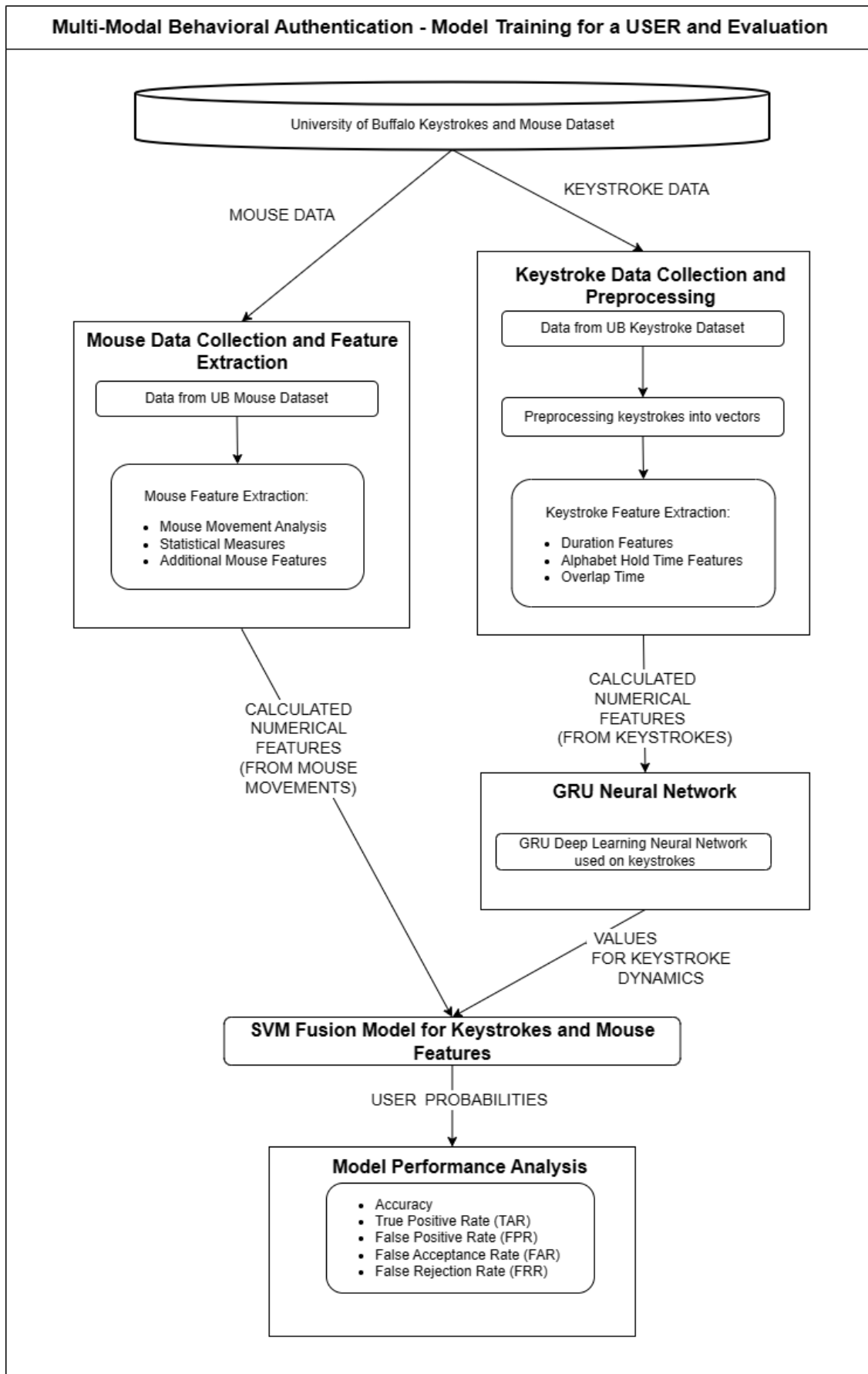


FIGURE 3.1: Architecture diagram for Training the Model based on a User's Keystrokes and Mouse movements

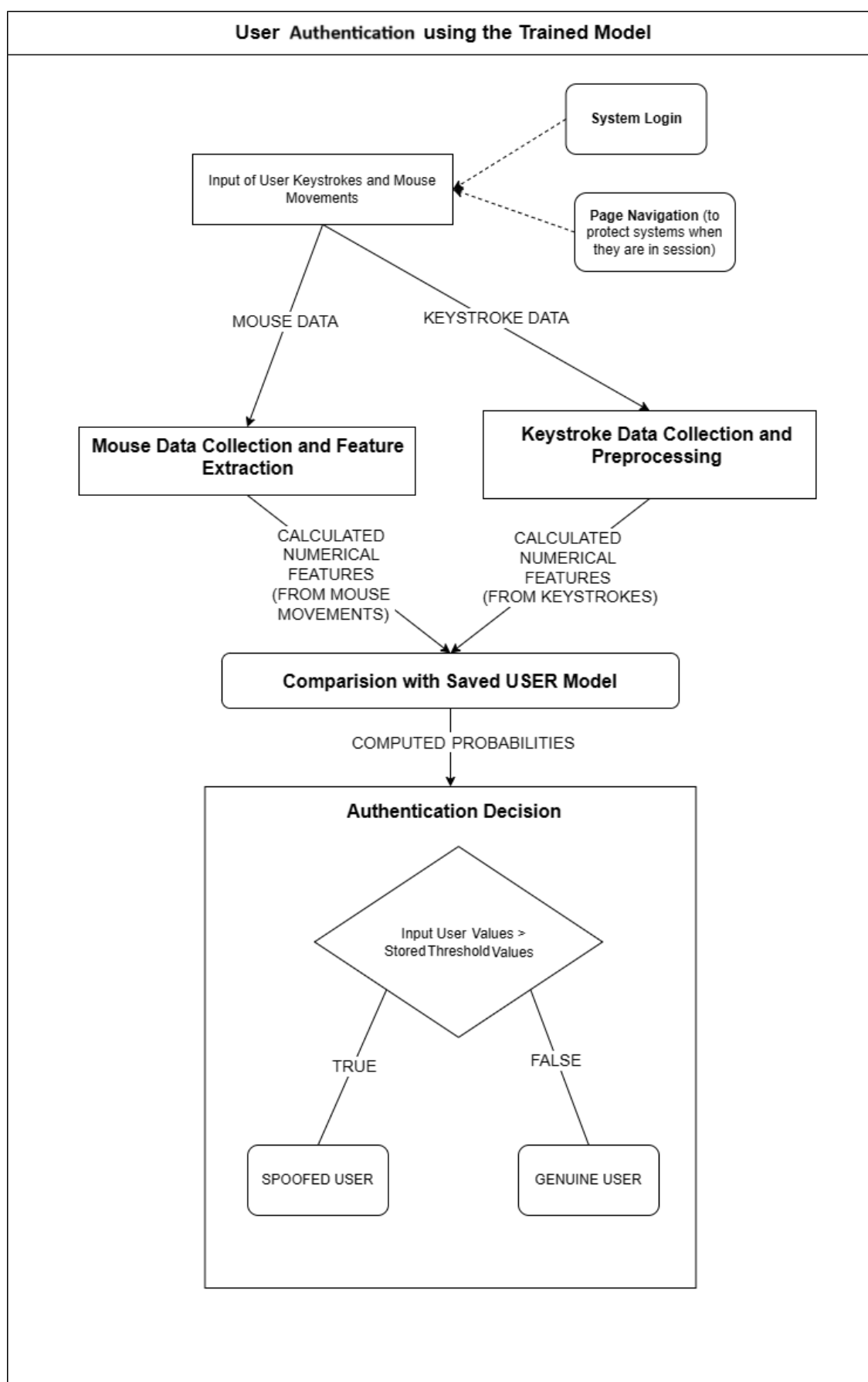


FIGURE 3.2: Architecture diagram for Using the saved model for performing authentication

3.1 USER REGISTRATION

The user registration component of the system, as indicated by Algorithm 1, is responsible for collecting the keystrokes and mouse movements of an individual and processing them. This includes pre-processing the data into an appropriate format, extracting the relevant features using the Feature Extraction Module, and feeding them into the Machine Learning Architecture to train the virtual fingerprint of the given individual with the pre-processed data. Evaluation metrics like TAR, ARR, FAR, FRR and accuracy are used to analyse the performance of the model.

Algorithm 1 User Registration Algorithm

- 1: **Data Processing:**
 - 2: **For** each user u :
 - 3: $K_u \leftarrow \text{ExtractKeystrokeDataFeatures}(u)$
 - 4: $M_u \leftarrow \text{ExtractMouseDataFeatures}(u)$
 - 5: **End for**
 - 6:
 - 7: **Model Training:**
 - 8: Initialize parameters: W_r, W_z, W_h, B_0, a_i
 - 9: Initialize hidden state: h_0
 - 10: **For** each user u :
 - 11: $h_u \leftarrow \text{GRU}(K_u, W_r, W_z, W_h, h_0)$ – GRU component for Keystrokes
 - 12: $f(u) \leftarrow B_0 + \sum_i a_i \cdot (M_u, \mathbf{x}_i)$ – SVM Fusion component for Mouse Movements
 - 13: **End for**
 - 14:
 - 15: **Model Evaluation:**
 - 16: Split data into training and validation sets
 - 17: Compute accuracy, TAR, TRR, FAR, FRR, and EER
 - 18: Log final evaluation results
-

3.1.1 Data Processing

The first step in user registration is the collection and analysis of keystroke and mouse data features for behavioral authentication. For keystroke data, fixed-length sequences are generated to capture key press and release timings,

facilitating subsequent processing. Feature extraction encompasses various parameters such as duration features between key presses, and key hold features and overlapping keys, when it comes to keystrokes, providing insights into user typing behavior. Similarly, for mouse data, features are extracted to capture spatial characteristics and movement patterns, including distances between consecutive mouse positions, angles of mouse movements, and statistical measures computation such as mean and standard deviation.

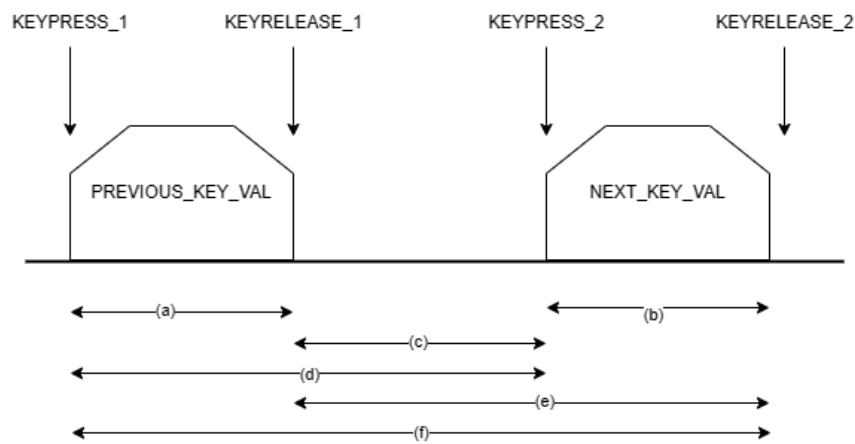


FIGURE 3.3: Data collected from each Keystroke Pair

3.1.1.1 Keystroke Data Features

1. **Pre-processing of Keystrokes:** The keystroke data is first converted into fixed-length sequences (with a common sequence length 16 used in many authentication scenarios) to facilitate further processing. Each sequence consists of consecutive keystrokes within a specified window size. Figure 3.3 represents the following:

(a) Hold duration of previous key

(b) Hold duration of next key

- (c) Time elapsed between the release of the previous key and the press of the next key.
- (d) Time elapsed between the press of the previous key and the press of the next key.
- (e) Time elapsed between the release of the previous key and the release of the next key.
- (f) Time elapsed between the press of the previous key and the release of the next key.

2. **Feature Extraction:** The extracted keystroke features from user typing behavior include the following:.

(a) **Duration Features:**

- *Minimum duration:* The shortest duration among all key presses and releases within a sequence.
- *Percentiles:* Statistical measures including the 25th, 50th (median), and 75th percentiles of key press and release durations, capturing the variability in timing.
- *Maximum duration:* The longest duration among all key presses and releases within a sequence.

(b) **Alphabet Hold Features:**

- *Mean alphabet hold duration:* The average duration of holding each alphabet letter key.
- *Standard deviation of alphabet hold duration:* The measure of variability in the duration of holding each alphabet letter key.

(c) **Overlap Time Features:**

- *Mean overlap time*: The average duration of simultaneous key presses or releases.
- *Standard deviation of overlap time*: The measure of variability in simultaneous key event durations.

3.1.1.2 Mouse Data Features

The extracted mouse features collectively provide a comprehensive representation of mouse interaction behavior, capturing both spatial characteristics and movement patterns.

1. Mouse Movement Sequences Analysis:

- Distances between Consecutive Mouse Positions (Euclidean distance between two consecutive mouse positions):
 - (a) *Distance variance*: The variability or spread of distances between consecutive mouse positions.
- Angles of Mouse Movements:
 - (a) *Calculation of angles*: The angle formed by the trajectory of mouse movement between two consecutive positions.
 - (b) *Angle variance*: The variability or dispersion of angles between consecutive mouse movements.

2. Statistical Measures Computation:

- Mean and Standard Deviation:

- (a) *Mean distance*: The average distance between consecutive mouse positions.
- (b) *Mean angle*: The average angle of mouse movement between consecutive positions.
- (c) *Standard deviation of distance*: The measure of spread or variability in distances.
- (d) *Standard deviation of angle*: The measure of variability or dispersion in angles.

3.1.2 Model Training

The model architecture employed for the problem is a combination of Gated Recurrent Unit (GRU) and Support Vector Machine (SVM). GRU is a recurrent neural network that is able to capture temporal dependencies in sequential data like keystroke dynamics. It utilizes gating mechanisms to regulate information flow. The GRU deep learning model integrates multiple layers of GRUs and dense layers for classification, with various optimization techniques and callbacks for efficient training. The SVM component of the model architecture integrates mouse movement data alongside keystroke features for registering the *virtual fingerprint*. Statistical analysis is employed to extract relevant features, and both models are configured with specific parameters for optimal performance.

3.1.2.1 GRU - Gated Recurrent Unit

A Gated Recurrent Unit (GRU) is a type of recurrent neural network (RNN) architecture that is particularly effective for capturing temporal dependencies in sequential data. GRU is designed to address some of the limitations of traditional RNNs, such as the vanishing gradient problem, which can hinder the learning of long-term dependencies in sequential data.

GRU achieves this by introducing gating mechanisms that regulate the flow of information within the network. These gating mechanisms include an update gate and a reset gate, which control how much past information to retain and how much new information to incorporate at each time step. By adaptively updating the hidden state based on the input and the previous hidden state, GRU can effectively capture and store relevant temporal information over long sequences.

One of the key advantages of GRU is its ability to learn and adapt to varying lengths of input sequences, making it well-suited for modeling sequential data with irregular or variable-length patterns. Additionally, GRU has fewer parameters compared to other RNN variants like Long Short-Term Memory (LSTM), making it computationally more efficient and easier to train, especially on smaller datasets.

In the context of keystroke dynamics modeling, GRU is beneficial for capturing the temporal patterns and dependencies inherent in the sequence of keystrokes. It can learn the sequential relationships between keystrokes, such as the timing and order of key presses, and effectively encode this information into a fixed-size representation, which can then be used for user authentication. By leveraging the temporal capabilities of GRU, our model can discern subtle variations and

nuances in users' typing behavior, enhancing the accuracy and robustness of the authentication system.

GRU - Mathematical Representation

The equations used to calculate the reset gate, update gate, and hidden state of a GRU are as follows

$$\text{Reset gate: } r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (3.1)$$

$$\text{Update gate: } z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (3.2)$$

$$\text{Candidate hidden state: } h'_t = \tanh(W_h \cdot [r_t \cdot h_{t-1}, x_t]) \quad (3.3)$$

$$\text{Hidden state: } h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot h'_t \quad (3.4)$$

Where:

h_{t-1} represents the previous hidden state.

x_t denotes the input at time step t .

W_r, W_z , and W_h are weight matrices.

σ represents the sigmoid activation function.

$\tanh$ denotes the hyperbolic tangent function.

GRU Deep Learning Model for Keystroke Data

- The GRU model architecture, whose parameter information is mentioned in Table 3.1, comprises multiple layers of Gated Recurrent Units (GRUs) followed by dense layers for classification tasks.
- Adam optimizer with a low learning rate is utilized to train the model. The choice of optimizer helps in efficiently updating model weights during training to minimize the loss function.
- During training, a validation split is employed to evaluate the model's performance on a separate validation set. This aids in monitoring the model's generalization ability and prevents overfitting.
- To achieve better convergence, early stopping is implemented as a callback mechanism. The training process halts if the validation loss does not improve after a certain number of epochs (patience parameter), thereby preventing unnecessary training and saving computational resources.
- A learning rate scheduler is incorporated to adjust the learning rate during training. The learning rate is decayed by 5% every 20 epochs, allowing the model to converge more effectively towards the optimal solution.
- A custom callback function is implemented to log the testing accuracy after each epoch. This callback writes the epoch number and testing accuracy to a CSV file for later analysis and monitoring of model performance.

TABLE 3.1: Model Parameters of GRU Component for Keystrokes

GRU Parameter	Value
Input Layers	128 - to Receive and process data.
Activation Functions	ReLU, Sigmoid - Add non-linearity.
Normalization	BatchNormalization - Stabilize and accelerate training.
Dropout	0.5 - Prevent overfitting.
Recurrent Layers	GRU - Handle sequence data.
Pooling	GlobalMaxPooling1D - Aggregate information.
Optimizer	Adam optimizer - Update model weights.
Loss Function	Binary Crossentropy - Measure prediction accuracy.
Training	500 epochs, batch size 128 - Learn patterns from data.
Callbacks	EarlyStopping, LearningRateScheduler - Control training process, so that if loss doesn't improve after certain epochs, it doesn't run all the way up to 500, thus preventing unnecessary training.
Training Data	Keystrokes Features

3.1.2.2 SVM - Support Vector Machine

Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression tasks. It aims to find the optimal hyperplane that best separates different classes in a high-dimensional space.

SVM works by maximizing the margin, which is the distance between the hyperplane and the closest data points (support vectors) of each class. It is effective in handling both linearly separable and non-linearly separable data through the use of kernel functions.

SVM offers robustness against overfitting, especially in high-dimensional spaces, due to its ability to find the optimal separating hyperplane.

In the context of mouse dynamics modeling, SVM can effectively classify and distinguish between genuine user movements and impostor movements. By learning the complex decision boundaries between different classes of mouse movements, SVM enhances the accuracy and reliability of authentication systems. SVM's ability to handle multi-dimensional feature spaces and its flexibility in handling non-linear relationships make it suitable for capturing intricate patterns in mouse dynamics. It provides a clear decision boundary, making it interpretable and suitable for real-time applications.

SVM - Mathematical Representation

For SVM, the prediction for a new input $\mathbf{x}$ is calculated using the dot product between $\mathbf{x}$ and each support vector $\mathbf{x}_i$:

$$f(\mathbf{x}) = B_0 + \sum_i a_i \cdot (\mathbf{x}, \mathbf{x}_i) \quad (3.5)$$

Here:

- B_0 represents the bias term.
- a_i are the Lagrange multipliers associated with the support vectors.
- $(\mathbf{x}, \mathbf{x}_i)$ denotes the dot product between the input $\mathbf{x}$ and the support vector $\mathbf{x}_i$.

SVM Fusion for Keystrokes and Mouse Features

- Support Vector Machine (SVM) model, whose parameter information is mentioned in Table 3.2, is utilized to incorporate mouse movement data into the model training process. Some Keystroke features are also integrated to emphasize their effect in the SVM model.
- Mouse movement sequences are analyzed to extract various features that provide insights into user behavior.
 1. *Statistical Analysis of Movement Angle Variations:* This category encompasses statistical metrics related to variations in the angle of mouse movement. These metrics are computed over varying intervals between successive mouse points. By examining these variations, insights into the trajectory and directional changes of mouse movements can be obtained.
 2. *Statistical Analysis of Movement Distance and Speed:* Within this category, statistical values regarding the distance covered and speed of mouse movement across different directional sectors are considered. These values are computed over varying intervals between mouse points. By examining the distance traveled and speed of movement within various directional sectors, a comprehensive understanding of mouse behavior and interaction patterns can be achieved.
- Statistical measures such as mean and standard deviation are computed for these features to capture the patterns and characteristics of mouse behavior effectively.

- SVM Model outputs the final output probabilities. These output values are used for indicating the likelihood of a user being an imposter (the user who claims to be the owner of the credentials but actually does not own them).

TABLE 3.2: Model Parameters of SVM Fusion Component for Mouse Movements

SVM Parameter	Value
Model Type	Support Vector Machine (SVM)
Kernel	Radial Basis Function (RBF)
Gamma	'scale' (default value)
C	Regularization parameter set to 1.0
Probability	Enabled to obtain class probabilities
Feature Scaling	StandardScaler used for feature normalization
Training Data	Combined features extracted from keystrokes and mouse dynamics

3.1.3 Model Evaluation

3.1.3.1 Model Validation and Evaluation Criteria:

In the implemented model, validation and evaluation are crucial steps to ensure the robustness and generalization of the user authentication system. Separate validation and test datasets are utilized for this purpose. The following Evaluation metrics are computed to assess the model's effectiveness:

- Accuracy
- True Positive Rate (TAR)

- False Positive Rate (FPR)
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)
- Equal Error Rate (EER)

These metrics offer insights into various aspects of the model's performance, such as its ability to accurately distinguish between genuine and impostor behavior, its capacity to detect anomalies, and its overall robustness in real-world scenarios. The validation and evaluation process ensures that the model's performance meets the desired criteria for reliable authentication in diverse usage environments.

3.1.3.2 Logging Final Results:

Final evaluation results, including accuracy and performance metrics such as true positive rate (TAR), false positive rate (FPR), false acceptance rate (FAR), and false rejection rate (FRR), are logged to a designated file for further analysis and comparison. Figure 3.4 demonstrates a graphical representation of the accuracy observed in training given the number of epochs.

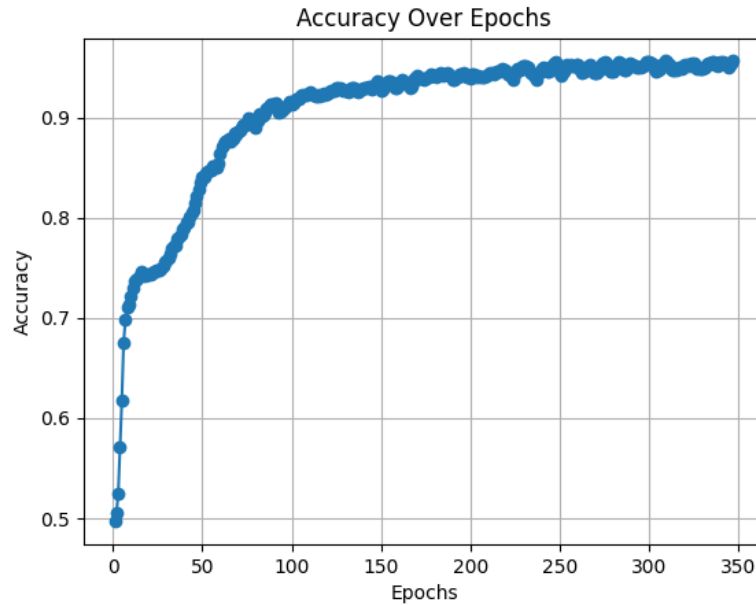


FIGURE 3.4: Accuracy increase over higher epochs of training

3.2 USER AUTHENTICATION

After registering the virtual fingerprint for the user, the user authentication component of the system, as indicated by Algorithm 2, uses the saved model to compare against the input keystrokes and mouse movements. The authentication process requires the same level of pre-processing and feature extraction to input to the saved user model in order to output the imposter probabilities. The output probabilities are then compared with a threshold, and if the probabilities exceed the threshold, an appropriate decision is made based on whether the data is from a genuine user or an imposter.

Algorithm 2 User Authentication

```

1: procedure USERAUTHENTICATION
2:   KeystrokeData  $\leftarrow$  CollectKeystrokeData()
3:   MouseData  $\leftarrow$  CollectMouseData()
4:   PreProcessedData  $\leftarrow$  PreProcessData(KeystrokeData, MouseData)
5:   KeystrokeFeatures  $\leftarrow$  ExtractKeystrokeFeatures(PreProcessedData)
6:   MouseFeatures  $\leftarrow$  ExtractMouseFeatures(PreProcessedData)
7:   UserModel  $\leftarrow$  LoadSavedUserModel()
8:   KeystrokeProbabilities  $\leftarrow$  UserModel.Predict(KeystrokeFeatures)
9:   MouseProbabilities  $\leftarrow$  UserModel.Predict(MouseFeatures)
10:  OverallProbabilities  $\leftarrow$  Combine(KeystrokeProbabilities, MouseProbabilities)
11:  Decision  $\leftarrow$  CompareWithThreshold(OverallProbabilities)
12:  if Decision > Threshold then
13:    return Imposter
14:  else
15:    return Genuine
16:  end if
17: end procedure

```

The Integration of the model for User Authentication contains the following Steps:-

3.2.1 Input Data Collection

The input data collection happens during both login and page navigation events, to simulate a continuous verification process which simultaneously does not take

up continuous processing power every single time a user keystroke or mouse coordinate is collected.

3.2.2 Feature Extraction

The data from keystrokes and mouse movements are pre-processed into the same appropriate format and the relevant features pertaining to keystrokes and mouse movements are extracted to input to the model prediction stage of the pipeline.

3.2.3 Model Prediction

The pre-existing saved model of the user is utilized to input the extracted features. Corresponding output probability values are obtained which are further compared with a threshold.

3.2.4 Thresholding

A threshold probability is set for each registered user based on the user preferences, which can be modified by the administrator of the system.

3.2.5 Output Generation

The output probability for a user is compared with the set threshold. If the score exceeds the threshold, the behavior is classified as Imposter. Otherwise it is

classified as a genuine one. The administrator can also set a maximum imposter limit, such that if the results get classified as imposter for more than the set number of times, the user's credentials are 'Blocked', leading to being unable to use the account unless the administrator unblocks it.

CHAPTER 4

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

In the chapter, we present the empirical validation of our proposed methodology and evaluate its performance against predefined metrics. Organized into four sections, namely Dataset Description, Ecosystem, Experiments Conducted, and Performance Analysis, this chapter provides a comprehensive overview of our research process. Firstly, we detail the characteristics and sources of the datasets used, along with any preprocessing steps undertaken. Subsequently, we describe the hardware and software ecosystem utilized, followed by a breakdown of the experimental setup and methodologies employed. Finally, we delve into a thorough analysis of the experimental results, examining key performance metrics and drawing meaningful conclusions to guide future research and practical implementations.

4.1 DATASET DESCRIPTION

4.1.1 Dataset used

The benchmark dataset utilized here is the University of Buffalo (UB) Dataset for Keystroke Dynamics and Mouse Movements. The dataset contains keystrokes based on fixed-text transcription. A part of the dataset is generated using different

types of keyboards across sessions. In addition, the mouse coordinate data and related events data are also made available with the keystroke dataset.

4.1.2 Dataset Format

The keystroke data, a sample of which is shown in Figure 4.1, is stored in text files, with each line representing a keystroke event: Each singular event contains the following details –

- The Pressed Key
- Event Type (KeyUp/KeyDown)
- Timestamp

Researchers from SUNY Buffalo curated the Buffalo dataset, comprising extensive fixed-text keystroke data collected from 157 proficient keyboard users. This dataset encompasses keystroke information obtained during both fixed text inputs and responses to open-ended questions. Each participant engaged in three sessions of inputting, with an average of 5,700 keystrokes per session. Consequently, the cumulative average keystrokes across the three sessions surpassed 17,000. Notably, some of the 157 participants utilized keyboards with varying key layouts for text input [29].

There are two subsets of users divided based on the keyboards they use: **BASELINE** subset (ID from 001 to 075), with 75 users using the same type of keyboard across 3 sessions; and **ROTATION** subset (ID from 075 to 148), with

73 users using 3 different types of keyboard across 3 sessions. With respect to file organization - there is one folder for each session. Under the session folder, there are 2 folders named baseline subset and rotation subset. Under the subset folder there are files for each subject which includes the subject ID.

```
D: > FYP > FYPcoding > UB_keystroke_dataset > s0 > baseline > 005001.txt
1  LShiftKey KeyDown 63578866662708
2  O KeyDown 63578866664049
3  LShiftKey KeyUp 63578866664096
4  N KeyDown 63578866664143
5  O KeyUp 63578866664159
6  E KeyDown 63578866664237
7  N KeyUp 63578866664237
8  E KeyUp 63578866664315
9  Space KeyDown 63578866664330
10 Space KeyUp 63578866664408
11 O KeyDown 63578866664455
12 F KeyDown 63578866664502
13 O KeyUp 63578866664533
```

FIGURE 4.1: Sample Instance of Keystroke Data from the UB Dataset

```
D: > FYP > FYPcoding > UB_mouse_and_events_dataset > 002200.txt
1  license Stuff
2  Initialised at | 11/23/2015 10:53:44 AM
3  Computer Name | raycharles
4  System User Name | raycharles\truther
5  User does not belong to Administrative Group
6  IP Address 0 | 128.205.50.23
7  635838728242361662 | 11/23/2015 10:53:44 AM | All Process Hooks except the Mouse
8  Logging Started for the First time
9  =====
10 635838728243765665 | 11/23/2015 10:53:44 AM | Window Activated | C:\loggers\Acti
11 635838728244077665 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1128,623
12 635838728244077665 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1121,621
13 635838728244389666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1114,619
14 635838728244545666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1107,616
15 635838728244545666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1092,606
16 635838728244701666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1085,602
17 635838728244701666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1079,598
18 635838728244701666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1072,592
19 635838728244701666 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1064,588
20 635838728244857667 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1057,582
21 635838728244857667 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1049,576
22 635838728245013667 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1039,572
23 635838728245013667 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1030,566
24 635838728245325667 | 11/23/2015 10:53:44 AM | Mouse Coordinates | 1022,561
```

FIGURE 4.2: Sample Instance of Mouse-Movement Data from the UB Dataset

Mouse data, a sample of which is shown in Figure 4.2, is stored in text files, with each line representing a mouse event. Each mouse event includes the following details:

- Timestamp
- Date and Time
- Event Type (Mouse Coordinates/Clicks/Window Activated/Logging/etc.)
- Event Details (The actual coordinates, activated window, etc)
- Additional Info

These datasets are processed to extract relevant features for subsequent model training and evaluation.

4.2 ECOSYSTEM

The ecosystem section is structured into two main subsections: hardware and software. The hardware subsection provides an overview of the experimental setup's physical components, including details such as the CPU, GPU, RAM, and storage specifications. On the other hand, the software subsection delves into the software environment used in the project, covering aspects such as the operating system, programming languages, frameworks, and development tools employed during experimentation. Additionally, the subsection on programming languages and frameworks further expands into subsubsections detailing specific tools and

libraries utilized for various tasks, such as data preprocessing, feature extraction, model training, and evaluation.

4.2.1 Hardware

The experimental setup utilized a laptop enabled with GPU, featuring an Intel Core i5 processor clocked at 2.5 GHz and 8 GB of RAM.

4.2.2 Software

4.2.2.1 Operating System

The experiments were conducted on a machine running Windows 11 64-bit OS, chosen for its user-friendly interface and compatibility with a wide range of software applications.

4.2.2.2 Programming Languages and Frameworks

- **Python:** The primary programming language employed for development.
- **TensorFlow:** Utilized as the core framework for building and training deep learning models.
- **Scikit-learn:** Utilized for traditional machine learning tasks.
- **NumPy/Pandas:** Used for numerical computations and data manipulation.
- **Matplotlib:** Utilized for data visualization.

- **Python Flask:** Implemented with HTML and vanilla JavaScript for frontend data processing.
- **Visual Studio Code (VSCode):** Used as the primary integrated development environment (IDE) for coding and project management.

4.3 EXPERIMENTS CONDUCTED

The experiments conducted to validate our proposed methodology involved two distinct lines of implementation.

1. Firstly, we implemented our authentication model using Python, TensorFlow, and Keras libraries, leveraging machine learning techniques such as GRU (Gated Recurrent Unit) and SVM (Support Vector Machine). To ensure robust evaluation, we adopted a stratified k-fold cross-validation approach using the `StratifiedKFold` function from the `sklearn.model_selection` module in Python. This technique partitions the dataset into k equally sized folds while preserving the class distribution in each fold. The model was trained and tested iteratively on different combinations of training and testing data, allowing for a comprehensive assessment of its performance. The benchmark dataset used for evaluation was the UB dataset, known for its extensive keystroke dynamics data, enabling a thorough examination of the model's accuracy and effectiveness in user authentication scenarios.
2. Secondly, we developed a Python Flask-based sample Email clone application that integrated the functionalities of our proposed authentication

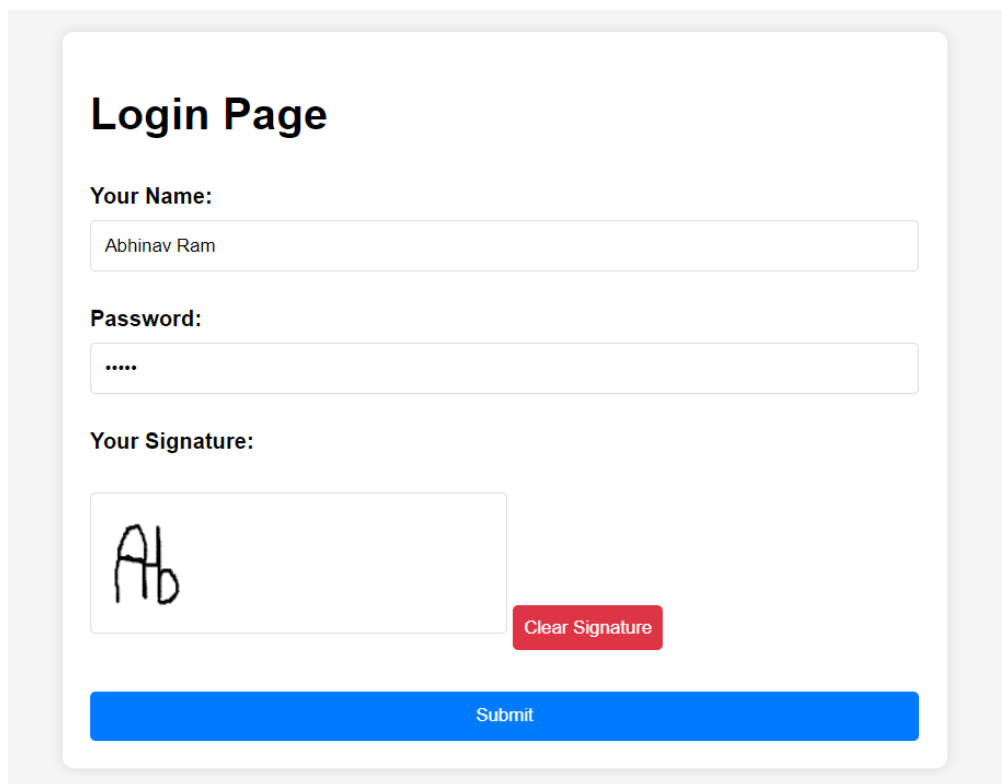
system. This application served as a platform to collect user keystroke and mouse data in real-time. Subsequently, the collected data was fed into the implemented GRU-SVM model, and the model was trained to recognize individual user behavior patterns. Additionally, a detection functionality was incorporated into the website to compare the current typing pattern with the saved model, enabling real-time authentication and validation of user identities.

Through these experiments, we aimed to assess the effectiveness and robustness of our proposed methodology in practical authentication scenarios, thereby contributing to the advancement of secure user authentication technologies.

4.3.1 User Interface

The proposed design was tested using a sample user interface, which comprises a website demonstrating the functionalities of the developed system. The website allows users to perform the following actions:

- **User Authentication:** The website implements a user authentication system requiring users to log in with their credentials, consisting of a username, password and signature, as shown in Figure 4.3. Upon successful authentication, the user's session is activated, granting access to the system's features. To maintain session integrity and security, multiple simultaneous logins for the same user are prevented.



The image shows a login page titled "Login Page". It contains three input fields: "Your Name:" with the text "Abhinav Ram", "Password:" with masked characters ".....", and "Your Signature:" with a handwritten signature "Ab". To the right of the signature field is a red button labeled "Clear Signature". At the bottom is a large blue button labeled "Submit".

FIGURE 4.3: User logs in to the system using required credentials

- **Data Collection:** The website provides an interface for users to input text or interact with the system, capturing their keystrokes, as shown in Figure 4.4, and mouse movements, as shown in Figure 4.5 in real-time. This data collection process is essential for generating user-specific behavioral biometric profiles used in authentication.

```
m KeyDown 1711728065630
m KeyUp 1711728065736
Shift KeyDown 1711728065789
a KeyDown 1711728065979
a KeyUp 1711728066108
t KeyDown 1711728066231
t KeyUp 1711728066323
t KeyDown 1711728066395
t KeyUp 1711728066508
h KeyDown 1711728066589
h KeyUp 1711728066729
e KeyDown 1711728066743
w KeyDown 1711728066926
e KeyUp 1711728067000
Shift KeyUp 1711728067058
w KeyUp 1711728067110
Space KeyDown 1711728067383
Space KeyUp 1711728067553
j KeyDown 1711728067717
j KeyUp 1711728067850
Shift KeyDown 1711728067896
o KeyDown 1711728068079
```

FIGURE 4.4: Sample showing the keystrokes data collected

1711728073809	29/3/2024, 9:31:13 pm	Mouse Coordinates	504,459
1711728073815	29/3/2024, 9:31:13 pm	Mouse Coordinates	504,456
1711728073829	29/3/2024, 9:31:13 pm	Mouse Coordinates	503,450
1711728073845	29/3/2024, 9:31:13 pm	Mouse Coordinates	501,440
1711728073862	29/3/2024, 9:31:13 pm	Mouse Coordinates	501,434
1711728073879	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,426
1711728073895	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,419
1711728073913	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,415
1711728073928	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,411
1711728073945	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,410
1711728073962	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,409
1711728073978	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,406
1711728073996	29/3/2024, 9:31:13 pm	Mouse Coordinates	500,405
1711728074012	29/3/2024, 9:31:14 pm	Mouse Coordinates	500,403
1711728074028	29/3/2024, 9:31:14 pm	Mouse Coordinates	500,402
1711728074045	29/3/2024, 9:31:14 pm	Mouse Coordinates	500,402
1711728074111	29/3/2024, 9:31:14 pm	Mouse Coordinates	500,403
1711728074129	29/3/2024, 9:31:14 pm	Mouse Coordinates	504,411
1711728074145	29/3/2024, 9:31:14 pm	Mouse Coordinates	505,414
1711728074163	29/3/2024, 9:31:14 pm	Mouse Coordinates	507,420
1711728074179	29/3/2024, 9:31:14 pm	Mouse Coordinates	507,422
1711728074197	29/3/2024, 9:31:14 pm	Mouse Coordinates	507,423

FIGURE 4.5: Sample showing the mouse movements data collected

- **Model Training:** We collected manual data encompassing keystrokes and mouse movements. The raw dataset includes over 1900 keystrokes and 1200 mouse movements for each user. This dataset was organized into 20 transitions, where each transition represented either a page login or a page transition. After collecting user input data, it undergoes processing through a pre-trained machine learning model. A combination of genuine user data and imposter users' data are input to the model to ensure discriminability. This model evaluates the input data against established patterns of user behavior to determine the likelihood of impostor activity.
- **Model Testing:** The testing of Genuine/Imposter happens both at login and at page navigation. Administrators can configure a threshold probability to balance between false positives and false negatives. Additionally, the system tracks the genuine/impostor detections for each user, as shown in Figure 4.6, and imposes restrictions, such as locking the user out after a certain number

of consecutive imposter detections, as shown in Figure 4.7. This proactive approach helps mitigate the risk of unauthorized access by identifying and responding to potentially malicious behavior.



```

result_log.txt x
result_log.txt
1
2
3 The Imposter probability threshold has been reached.
4 Imposter activity is detected! User 'Abhinav Ram' has been blocked
5 Sending email to admin...
6
7 ('Activity within normal limits for User "Abhinav Ram" (Anomaly Probability =', 31.06, '%)')
8 ('Slight Suspicion of Anomalous Activity in User "Abhinav Ram" (Anomaly Probability =', 71.32, '%)')
9 ('Slight Suspicion of Anomalous Activity in User "Abhinav Ram" (Anomaly Probability =', 72.7, '%)')
10 ('Slight Suspicion of Anomalous Activity in User "Abhinav Ram" (Anomaly Probability =', 74.49, '%)')
11
12
13 The Imposter probability threshold has been reached.
14 Imposter activity is detected! User 'Abhinav Ram' has been blocked
15 Sending email to admin...
16
17

```

FIGURE 4.6: Logging the Genuine/Imposter detections in a backend file

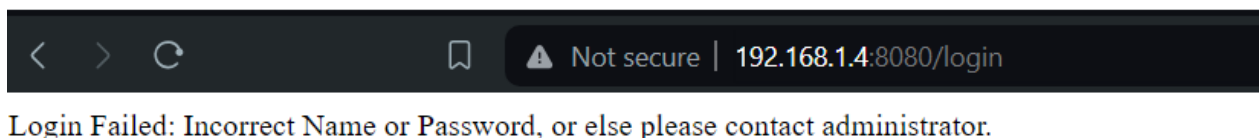


FIGURE 4.7: Blocks User after certain number of consecutive Imposter detections

- **Alert Notification:** In the event of suspicious activity detected by the model, an email alert is automatically generated and sent to designated recipients. The email notification includes details of the detected activity, enabling administrators or security personnel to take appropriate action. This prompt notification facilitates timely response to security incidents, minimizing the impact of potential breaches.

- **Message Sending:** Users can compose and send messages to other users within the system through the website's messaging functionality, as indicated in Figure 4.8b. These messages are securely stored and can only be accessed by the intended recipients, as indicated in Figure 4.8a. The sample mail/message sending application establishes an appropriate platform for testing the integrability of the data collection of the keystrokes and mouse movements.

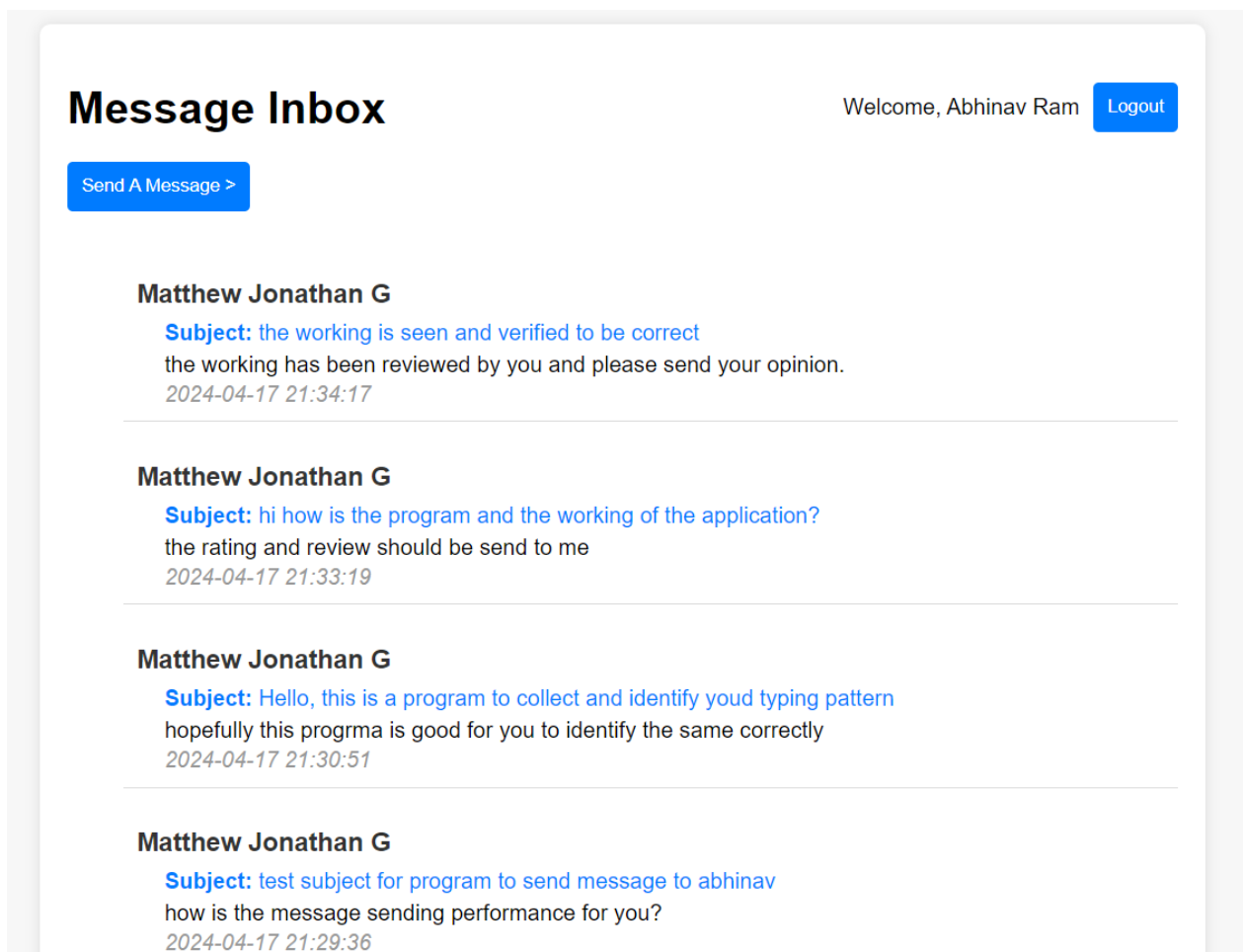


FIGURE 4.8A: Sample Mail Application: Inbox

Send Message Welcome, Abhinav Ram [Logout](#)

Receiver's Name:

Matthew Jonathan G

Subject:

Hello This is a Subject

Message Body:

Hello This is a body of the message. This is to demonstrate Keystroke and Mouse Authentication

Your Signature:

Ab

[Clear Signature](#)

[Submit](#)

[Back to Messages](#)

FIGURE 4.8B: Sample Mail Application: Sending a Message

4.4 PERFORMANCE ANALYSIS

The performance analysis section delves into a detailed evaluation of the implemented user authentication models. Through rigorous experimentation and assessment, the efficacy and robustness of the models are scrutinized across

various performance metrics. By dissecting the models' ability to discern between genuine users and imposters, this section provides valuable insights into their discriminatory power and generalization capabilities.

The chosen metrics for evaluation are accuracy, False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER). These metrics serve as key performance indicators for assessing the effectiveness of the authentication models. Below are the formulas for the chosen metrics:-

4.4.1 Performance analysis metrics

4.4.1.1 TPR (Sensitivity)

True Positive Rate (TPR) is also known as Sensitivity:

$$TPR = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

4.4.1.2 FPR (1 - Specificity)

False Positive Rate (FPR) is calculated as 1 minus Specificity:

$$FPR = 1 - \text{Specificity}$$

Where Specificity is also known as True Negative Rate (TNR):

$$TNR(\text{Specificity}) = 1 - \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

4.4.1.3 Equal Error Rate (EER)

Equal Error Rate (EER) is calculated as the average of False Positive Rate (FPR) and False Negative Rate (FNR), where FPR and FNR are closest:

$$EER = \frac{FPR + FNR}{2}$$

Below are the tables showcasing chosen performance metrics across the Baseline and Rotation subsets of the UB dataset. These metrics are obtained upon training 3000 entries (of both user and randomly selected imposter data each) of keystrokes and 3000 (of both user and randomly selected imposter data each) of mouse movements. Table 4.1 and 4.2 denote the cross validation results (computed for each fold and Table 4.3 represents the overall results for the baseline and rotation datasets)

TABLE 4.1: Results of Each Fold in Cross-Validation (Baseline)

No of Fold	TAR	TRR	FAR	FRR	Accuracy (in %)
1	0.492327	0.472244	0.027137	0.007271	96.4572
2	0.492374	0.472253	0.027119	0.007234	96.4627
3	0.492383	0.472318	0.027044	0.007234	96.4701
4	0.492364	0.472336	0.027045	0.007235	96.4701
5	0.492364	0.472279	0.027102	0.007235	96.4644

TABLE 4.2: Results of Each Fold in Cross-Validation (Rotation)

No of Fold	TAR	TRR	FAR	FRR	Accuracy (in %)
1	0.478981	0.479946	0.019835	0.020869	95.8928
2	0.478937	0.479944	0.019840	0.020912	95.8881
3	0.478885	0.479913	0.019868	0.020962	95.8799
4	0.478850	0.479860	0.019919	0.020998	95.8711
5	0.47880	0.479829	0.019947	0.021045	95.8637

TABLE 4.3: Performance of the proposed system on UB Dataset - baseline and rotation

Metrics	Baseline (in %)	Rotation (in %)
Accuracy	96.48	95.8
True Acceptance Rate (TAR)	49.23	47.90
True Rejection Rate (TRR)	47.24	48.00
False Acceptance Rate (FAR)	2.70	1.98
False Rejection Rate (FRR)	0.72	2.09
Equal Error Rate (EER)	1.71	2.03

Upon comparison with the other common Deep Learning (DL) Architecture used, we observed that this achieves a balance between training time and performance:

TABLE 4.4: Comparison with other Existing Approach

Model	FRR (in %)	FAR (in %)	EER (in %)
CNN + RNN [28]	4.27	1.39	2.83
Proposed GRU + SVM	2.09	1.98	2.03

The proposed system demonstrates a 39.41% improvement over the existing DL approach.

The Receiver Operating Characteristic (ROC) curve, as shown in Figure 4.9, is a plot of the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The TPR represents the proportion of true positives correctly identified, while the FPR represents the proportion of false positives incorrectly identified. The ROC curve is particularly useful for visualizing the trade-off between sensitivity and specificity.

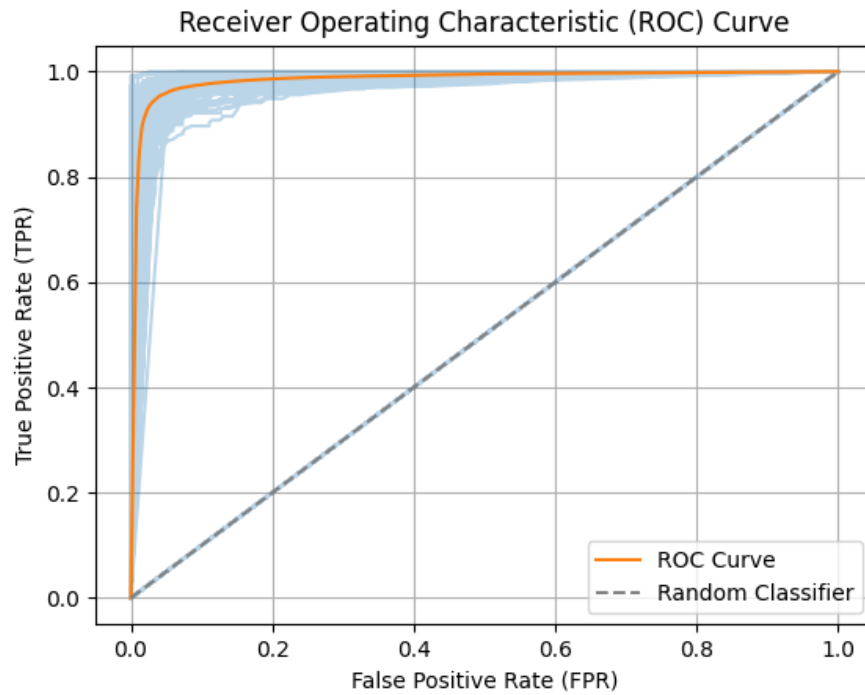


FIGURE 4.9: ROC Curve between TPR and FPR values obtained from the trained model

The individual light blue curves represent the TPR and FPR values for each user, where each point represents training with different threshold values, and the orange curve represents the average values of each threshold. A high area under the ROC curve indicates the discrimination performance of the model.

CHAPTER 5

SOCIAL IMPACT AND SUSTAINABILITY

Through the implementation of this project, we would have to incorporate the following considerations pertaining to society, health, safety, legal, environment, and culture.

1. Society:

- **Privacy Concerns:** The project involves the collection and analysis of keystroke and mouse data, raising concerns about user privacy and data security. Ensuring that adequate measures are in place to protect users' sensitive information and uphold their privacy rights is paramount.
- **Ethical Considerations:** Ethical considerations arise regarding the transparent and responsible use of keystroke and mouse data. It's essential to adhere to ethical guidelines and obtain informed consent from users regarding data collection and usage.
- **User Perception:** User perception of the project's objectives and methods is crucial. Educating users about the purpose and benefits of keystroke and mouse data analysis can help build trust and foster acceptance of the project.

2. Health and Safety:

- **Ergonomic Awareness:** The project may impact user health and safety, particularly concerning ergonomic keyboard and mouse usage. Promoting awareness of ergonomic practices and providing guidance

on healthy typing and mouse habits can mitigate the risk of repetitive strain injuries (RSI) and musculoskeletal disorders.

3. Legal:

- **User Consent:** Obtaining explicit consent from users for data collection and processing activities is a legal requirement. Clearly communicating the purposes and implications of data collection helps users make informed decisions about their participation in the project.

4. Environment:

- **Energy Efficiency:** The project's computational and hardware requirements may have implications for energy consumption and environmental sustainability. Implementing light weight models and energy-efficient algorithms can minimize the project's environmental footprint.

5. Culture:

- **Cultural Sensitivity:** The project should consider cultural diversity and sensitivity in its design and implementation. Ensuring that the project is inclusive and respectful of diverse cultural norms, languages, and user preferences promotes accessibility and user acceptance.
- **Inclusivity:** Designing the project to accommodate users with varying cultural backgrounds and technological literacy levels enhances its accessibility and usability, fostering inclusivity and participation across diverse user groups.

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

In this report, we have worked on the design and associated algorithms for a project aimed at Multimodal Behavioral Authentication for Keystrokes and Mouse movements. Leveraging a powerful model architecture, which includes a combination GRU (Gated Recurrent Unit) and SVM (Support Vector Machine), our project focused on developing robust authentication models capable of effectively distinguishing between genuine and impostor users. GRU, inspired by its efficacy in sequence modeling, served as a pivotal component in capturing temporal patterns within user interactions. Concurrently, SVM, with its versatility in handling complex data distributions, provided a powerful tool for classification tasks. By combining the two, we aimed to improve the security and reliability of authentication systems in real-world scenarios. Through this project, we have laid the foundation for further exploration and refinement of authentication systems, paving the way for enhanced security measures in modern computing environments.

Our analysis of the performance metrics on both versions of the UB dataset reveals the robustness and reliability of the GRU-SVM model in user authentication. The model exhibits high accuracy rates and balanced error rates across datasets, indicating its effectiveness in distinguishing between genuine users and impostors. With consistent acceptance and rejection rates, the model

demonstrates its reliability in correctly identifying users. Moreover, its low false acceptance and rejection rates ensure a high level of security, minimizing unauthorized access while avoiding wrongful denial of genuine users. The low Equal Error Rates (EER) underscore the model's capability to maintain a balance between false acceptance and rejection, highlighting its overall efficacy in user authentication based on keystrokes and mouse movements.

6.2 Future Work

1. **API Integration:** We aspire to develop our authentication system into a versatile API that can seamlessly integrate with various custom applications and platforms. This would enable developers to leverage our robust authentication algorithms within their own software solutions, thereby enhancing security and user experience across diverse domains.
2. **Addressing Security Concerns:** Given the paramount importance of security in authentication systems, we are committed to further enhancing the robustness and resilience of our models against potential security threats. This entails conducting rigorous security assessments, implementing advanced encryption techniques, and fortifying our system against adversarial attacks and data breaches.
3. **Interpretable AI:** We recognize the significance of interpretability in machine learning models, especially in critical domains such as user authentication. To this end, we plan to explore techniques for developing interpretable AI or explainable AI models that provide insights into the decision-making process of our authentication algorithms. By enhancing

transparency and understandability, we aim to foster trust and confidence among users and stakeholders.

4. **Improved System Architecture:** We aspire to elevate our authentication system's architecture to ensure scalability, flexibility, and enhanced security. This involves incorporating highly secure data storage principles, such as encryption-at-rest and in-transit, to safeguard sensitive user information from unauthorized access or breaches. Additionally, we aim to implement a robust access control mechanism that distinguishes between administrator and user, enabling administrators to manage system configurations and user privileges efficiently. Moreover, we plan to optimize the system's scalability by designing modular components that can be easily scaled horizontally or vertically to accommodate varying workloads and user demands. By prioritizing architectural enhancements, we aim to bolster the system's resilience against evolving security threats while ensuring seamless scalability and deployment across diverse environments.

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